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 SURVEY

Human Detection From Unmanned Aerial Vehicles' Images for Search and Rescue Missions: A State-of-the-Art Review

AHMAD A. BANY ABDELNABI¹ AND GHAITH RABADI¹

IST, School of Modeling, Simulation, and Training, University of Central Florida, Orlando, FL 32816, USA

Corresponding author: Ahmad A. Bany Abdelnabi (Ahmad.BanyAbdelnabi@ucf.edu)

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ABSTRACT Natural disasters continue to occur at an alarming rate impacting communities worldwide and requiring greater preparedness and response efforts. In the aftermath of disasters such as hurricanes and earthquakes, search and rescue (SAR) usually takes the highest priority on the response list. Utilizing recent advancements in technology, especially Unmanned Aerial Vehicles (UAVs) and Computer Vision in SAR operations has become more common due to their ability to quickly scan large disaster areas, identify and locate victims, and deliver first aid supplies especially in challenging environments. Researchers have leveraged machine learning (ML), particularly Convolutional Neural Networks, to enhance human detection accuracy. However, despite achieving excellent recall rates, the efficiency of these algorithms during actual SAR missions remains a critical consideration. The aim of this paper is to thoroughly examine the literature on human detection from UAV aerial images in SAR scenarios. This paper reviews the existing literature comprehensively and categorizes methods based on disaster type, environment, case study availability, and UAV system capabilities. It also reviews in detail the ML approaches used in the literature and compares them based on factors like image types, training datasets, model details, processing hardware, and evaluation methods. Moreover, it reviews and compares the available datasets according to their types, quantities, applications, and relevance to SAR operations. It also briefly covers state-of-the-art hardware for real-time onboard processing. Finally, we discuss the main challenges and recommendations for achieving fully autonomous UAV systems for human detection in SAR missions and offer recommendations for future research directions.

INDEX TERMS Computer vision, disaster response, search and rescue, unmanned aerial vehicles.

I. INTRODUCTION

The frequency and intensity of natural disasters are on the rise resulting in significant losses in property and lives [1]. In the aftermath of disasters, rapid search and rescue (SAR), is usually the very first activity conducted after a disaster to save lives and reduce human suffering. However, detecting and locating humans can be a true challenge for SAR teams [2]. SAR operations can also be costly due to necessary actions such as sending large numbers of rescuers, search dogs, and various types of ground and aerial vehicles to locate and rescue people.

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Unmanned aerial vehicles (UAVs) have become more commonly used in SAR missions for their ability to effectively and efficiently scan disaster sites and capture images and videos in which human casualties can be detected and located, enhancing the possibility of locating missing people and increasing survival rates [3]. While UAVs have improved SAR, this technology is still broadly dependent on the human eye to review thousands of images or lengthy videos to locate humans, resulting in inefficiency. The emergence of Computer Vision (CV) holds a lot of promise to tackle this problem.

CV is the scientific discipline that utilizes machines and artificial intelligence to classify or describe images or videos similarly to how human vision functions [4]. Particularly,

deep neural network-based CV has become a widely used machine learning tool [5]. Among the most extensively researched subjects in CV is object detection (OD), which is the process of recognizing various kinds of objects, such as people, cars, animals, or other identifiable objects in pictures [6], [7], [8].

Enhancing human identification during disasters can be achieved by leveraging technologies capable of rapidly and accurately processing numerous images. Consequently, deploying UAVs equipped with Deep Learning (DL) models becomes highly advantageous for SAR teams. These UAVs can efficiently survey larger disaster-stricken areas, outperforming traditional methods.

In recent years, extensive research has explored the feasibility of using UAVs in SAR operations. For instance, [9] proposed “DronAID,” a real-time drone system that detects humans during disasters using passive infrared sensors (PIR). Reference [10] introduced an onboard human detection system for identifying swimmers in open waters, leveraging the You Only Look Once (YOLO) DL model. The system demonstrated high accuracy in critical maritime situations. Additionally, [11] proposed an innovative Internet-of-Drones (IoD) system with blockchain technology, edge servers, and UAVs to enhance SAR missions. This system supports different human detection approaches while ensuring traceability, transparency, and security.

The aim of this paper is to thoroughly examine the literature of human detection from UAV images in SAR scenarios. Its key objectives are providing 1) comprehensive literature review that analyzes existing research, and categorizes the used methods based on disaster type, environment, case study availability, and UAV system capabilities. 2) technology overview that provides insights into the used technologies for human detection in SAR operations. 3) ML Approaches review with detailed discussions of these approaches used in the literature, and comparing factors like image types, training datasets, model details, processing hardware, and evaluation methods. 4) Available dataset comparisons based on to their types, quantities, applications, and relevance to SAR operations. 5) brief coverage of state-of-the-art hardware for real-time processing. Finally, the paper identifies and discusses the main challenges to achieving fully autonomous UAV systems for human detection in SAR missions and offers recommendations for future research directions.

The remainder of the article is organized as follows. Section II discusses the research methodology and how the literature was collected. Section III presents a general background and review of SAR operations. Section IV reviews the available CV algorithms for SAR missions. In section V, a review for the available datasets and processing technologies for SAR missions is presented. Section VI discusses on-board processing hardware, and Section VII concludes with the challenges and future direction in the implementation of CV in SAR missions.

II. RESEARCH METHODOLOGY

Figure 1 presents the methodological framework employed in this study. This framework encompasses three phases: an exhaustive search for relevant research articles, a qualitative analysis of the gathered articles, and an interpretation of the findings with suggestions for future research.

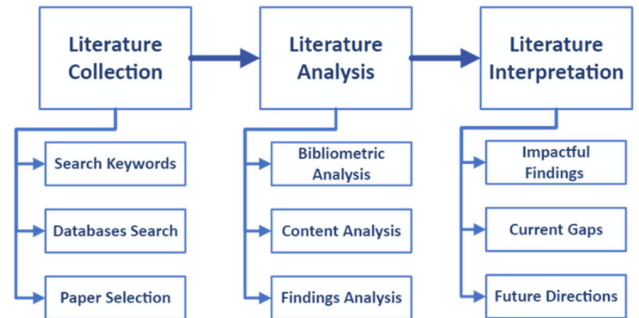


FIGURE 1. Research methodological framework.

A. LITERATURE COLLECTION

Google scholar, a leading academic search engine, was used to locate articles relevant to the topic. To minimize the likelihood of overlooking articles, additional databases such as “Scopus” and “Web of Science,” along with the websites of pertinent journals, were also examined. Three sets of search phrases were used, as depicted in Table 1, to ensure that the search results pertained to human detection, SAR missions, and UAVs. Every possible combination of these three search phrases, separated by either “AND” or “OR,” was used to search the selected database. The search was not confined to a specific time frame to encompass all relevant literature. An inclusive retrieval strategy was employed to obtain and shortlist the relevant articles.

TABLE 1. Search keywords used to collect the papers.

SAR related search terms	UAVs related search terms	CV related search terms
Search and Rescue	Unmanned Aerial Vehicle	Human/Person Detection
Post-Disaster Response	Drone	Computer/Machine Vision
Missing People	Aerial Images	Victim Identification
Victim		Machine Learning

Most retrieval strategies commonly used in literature reviews rely solely on authors’ keywords [12], [13], [14], yielding research articles that include the search phrases in their keywords. Although these techniques are widely accepted and used for retrieving research articles, in this study, the search fields were initially set to “Title, Abstract, Keywords” and later expanded to “All fields” to enhance inclusivity and reduce the chance of missing relevant articles.

This retrieval approach resulted in a large collection of articles, ensuring the inclusion of as many pertinent publications as possible. The authors then manually reviewed and filtered the collected articles, removing all book chapters, dissertations, reports, duplicates, unrelated files, and non-English resources, leaving only peer-reviewed academic conference and journal articles. Further filtration was performed on the selected articles to confirm their relevance, resulting in the selection of 99 papers for review. Figure 2 illustrates the literature collection and screening process.

The integration of object identification techniques with UAVs, particularly for SAR missions, is a relatively new topic in the research community. The evolution of this field over time can be observed by filtering the published articles by year of publication. Figure 3 shows the number of publications and the overall trend in the literature retrieved since 2012. There were only five results for this search until 2012. However, when articles to early 2024 are included, the number of results increases to 99. A significant shift occurred around 2020, marked by heightened interest in this interdisciplinary domain. The number of publications surged to 19 in 2020 and peaked at 22 in 2021. However, after 2021, there was a gradual decline in the number of publications, with only 17 projected for 2022.

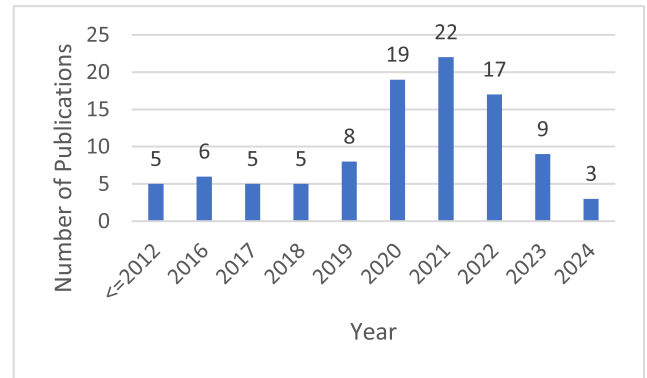


FIGURE 3. Yearly number of publications since 2012.

- **Manual Filtering:** The manual review process, while thorough, is subject to human error and subjective judgment. Implementing automated tools for initial screening could enhance the reliability of the selection process.

C. RELATED WORK AND RESEARCH CONTRIBUTIONS

Numerous reviews have been conducted on the applications of UAVs and the benefits they offer to various sectors, including traffic monitoring [15], remote sensing [16], and monitoring in mining locations [17]. A comprehensive study detailing a range of UAV uses, such as traffic monitoring, infrastructure inspection, product delivery, remote sensing, and search and rescue, is presented by [18] and [19]. Two other review studies, [20], [21], examine the significance of UAVs in wireless communications, discuss their advantages and disadvantages, and assess emerging markets. Reference [22] reviewed the factors affecting performance and significant obstacles to UAV use. The authors discussed the different types of UAVs, as well as the ethical, safety, confidentiality, and security issues associated with them. They then provided an overview of existing DL-based algorithms for detecting flying drones. Reference [23] analyzes the frequency, roles, pilots' abilities, and related possibilities and challenges in mountainous terrain related to the usage of rescue UAVs in Polish mountains across geographical divisions.

There are also some review articles about the use of UAVs in disaster response and SAR operations. For instance, [24] aim to assess the practicality of ongoing UAV projects and tackle a few of the challenges related to UAV deployment during major disasters, hoping to empower and motivate future work. Reference [25] provide a comprehensive summary of the crucial function that UAVs play in various SAR operations. They provide a brief overview of UAV types, components, and their significance in SAR operations. Furthermore, they offer an extensive analysis of sensor utilizations in UAVs for SAR missions, emphasizing their functions in object identification, localization, and sensation. However, the article is very general on the role of UAVs in SAR missions and does not focus on human detection.

With the significant advancement in CV and object detection in the last decade, there were also some review papers

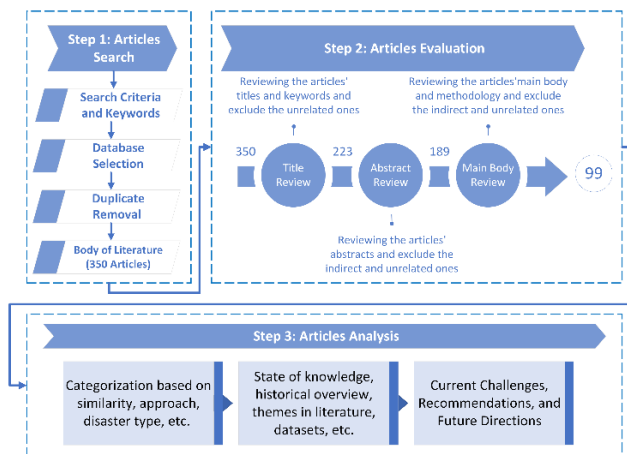


FIGURE 2. Literature collection and screening approach.

B. METHODOLOGICAL LIMITATIONS

To address potential biases and limitations in our methodology, we acknowledge the following points:

- **Selection Bias:** Despite our comprehensive search strategy, there is a possibility of selection bias due to the exclusion of non-English articles and grey literature. Future research should consider including these sources to provide a more holistic view.
- **Publication Bias:** The reliance on peer-reviewed articles may introduce publication bias, as studies with significant findings are more likely to be published. Including grey literature and unpublished studies could mitigate this bias.

TABLE 2. Summary of the contributions of existing surveys/reviews.

Article	Contribution	Difference	Common articles
[24]	evaluates the feasibility of ongoing drone projects and discusses some of the difficulties in using drones during major disasters with the hopes of inspiring and enabling more work in the future.	discusses the general use of UAVs in disaster management but does not focus on SAR (it only covers 12 SAR articles). It lacks detailed discussions about the ML approaches and datasets.	8 articles
[22]	examined the factors affecting drone performance and the significant obstacles to drone use. Discussed the different types of UAVs, as well as the ethical, safety, confidentiality, and security issues associated with them. provided an overview of existing DL-based algorithms for detecting flying drones.	focused on the UAVs and their safety and security issues, and the ML approaches focus on detecting the flying drones, not humans. However, it is not related to SAR missions.	None
[30]	A thorough examination of the most relevant developments, as well as the strengths and weaknesses of these developments, has been conducted in relation to the recently published uses and applications of DL for unmanned aerial vehicles. There's also an in-depth review of the main DL methods. They conclude by summarizing the main challenges in integrating DL with UAV-based technologies.	The focus is not on SAR, it is seven years old, and reviews less than 10% of the references discussed in our review.	7 articles
[23]	analyzes the frequency, roles, pilots' abilities, and related possibilities and challenges in mountainous terrain related to the usage of rescue drones in Polish mountains across geographical divisions.	It does not focus on human detection or ML. It focuses on the possibilities and limitations of using UAVs in SAR. Additionally, it focuses only on mountain SAR, and there are no details or discussions of ML approaches and datasets.	5 articles
[1]	presents a brief survey of the utilization of technology and UAVs in finding humans in SAR efforts in the aftermath of natural disasters. The article discusses the traditional approach and steps of finding people in disasters, and how UAVs can help in speeding up the process.	It is a short review about the types of UAVs and their applications in finding people in disasters. It only briefly mentions ML and CV. Additionally, there are no details or discussions about the ML approaches and datasets.	2 articles
[32]	discusses the state-of-the-art and obstacles in the area of DL-based OD tools for maritime UAVs., it covers the benefits and drawbacks of every method for marine object identification and offers a thorough analysis of them all.	focuses on detecting any type of objects in maritime such as ships, swimmers, etc. It is not about SAR operations, and it only focuses on maritime monitoring.	3 articles
[31]	It reviews the literature on detecting humans in maritime SAR.	It is a short review that focuses only on maritime SAR, and it is not focused on aerial/UAV images. Additionally, there are no details or discussions about the ML approaches and datasets.	1 article
[25]	provides a thorough summary of the important role that UAVs play in SAR missions. It provides a brief overview of UAV types, elements, and their significance in SAR operations. Furthermore, it offers an extensive analysis of sensor utilizations in UAVs, emphasizing their functions in target identification, localization, and sensation. It also goes into detail on the different ways that UAVs are being used in SAR, such as for on-site modeling and monitoring, target identification and localization, and search missions like task assignment, path planning, and collision avoidance. It also evaluates the merits and drawbacks of various approaches methodology employed in various studies.	It is very general on the role of UAVs in SAR missions and is not focused on human detection (it discusses it in a short subsection with few references). It focuses on 1) different types of UAVs and their sensors. 2) tasks other than human detection including wireless communications, path planning, Navigation, disaster area modeling, task assignment, and collision avoidance. Additionally, there are few details and discussion about the ML approaches for human detection and no details about the datasets.	13 articles
[33]	evaluates the current state-of-the-art in automating the assessment of disaster effect and using images from aerial vehicles to detect structural effects in collapsed constructions in order to extract useful information to help emergency responders. It also examined the different DL methods for processing the data.	It is a short review that focuses on assessing collapsed buildings for victim identification. It does not consider any other type of disasters, environments, or SAR missions.	1 article
[29]	analyzes the applied uses of the YOLO model on UAV images in engineering, logistics, farming, robotics, and other domains while presenting the development history of the integration between YOLO and UAVs.	reviews only the research on the YOLO model, and it is not about SAR operations. It reviews a general use of the YOLO model in detecting objects from UAV images such as face mask, vehicle, vegetables and fruits, concrete damage, and tree disease.	5 articles

about detecting humans using ML in different applications, for example, human face detection [26], activity recognition [27], and human emotion detection [28]. There are a few survey papers that review the literature on the implementation of different ML approaches to analyze UAV images. For instance, [29] analyzed the applied uses of the YOLO model on UAV images in engineering, logistics, farming, robotics, and other domains while presenting the development history of the integration between YOLO and UAVs. Reference [30] presented a thorough examination of the most relevant technologies, as well as the strengths and weaknesses of these technologies, in relation to the recently published uses and applications of DL for UAVs. They also provided an in-depth review of the main DL methods. They conclude by summarizing the main challenges in integrating DL with UAV-based technologies. However, this survey is not focused on SAR.

Few surveys and review studies on human detection from UAV images for SAR missions have been published in the past. The main limitations are that they are very short and cover a few numbers of studies, they are focused on one type of disaster or environment, and they have no details or discussions about the ML approaches and datasets. For example, [31] provide a short review of the literature on detecting humans in maritime SAR. However, it is a short review focused only on maritime SAR and it is not focused on aerial/UAV images. Additionally, there are no details or discussions of the ML approaches and datasets. Reference [32] discuss the state-of-the-art and obstacles in the area of DL-based OD methods for maritime UAVs. Additionally, it covers the benefits and drawbacks of the methods for maritime object identification and offers a thorough analysis of all of them. Nevertheless, the article focused on detecting any type of objects in maritime such as ships, swimmers, etc. It is not about SAR operations, and it only focuses on maritime monitoring.

Reference [33] evaluate the current literature in automating the assessment of disaster effect and using images from aerial vehicles to detect structural failure in collapsed constructions in order to extract useful information to help emergency responders. They also examined the different DL methods for processing the data. However, it is a short review that focused on assessing collapsed buildings for victim identification. It did not consider any other type of disasters, environments, or SAR missions. Finally, [1] presented a brief survey of the utilization of technology and UAVs in finding humans in SAR efforts in the aftermath of natural disasters. The article discusses the traditional approach and steps of finding people in disasters, and how UAVs can help in speeding up the process. However, it is a concise review of the types of UAVs and their applications in detecting people in disasters with one paragraph on ML and CV without much detail on ML approaches and datasets.

Table 2 provides an overview of existing survey contributions, highlights the distinctions between these surveys and our own, and enumerates the common papers shared among previous surveys and our survey.

The overall contribution of our work lies in its comprehensive examination of human detection based on DL methods from UAV images in SAR operations. The primary contributions of this review paper are as follows:

- 1) It offers a comprehensive, state-of-the-art, and in-depth review and analysis of the available literature on human detection from UAV images in SAR operations. It categorizes and compares them based on disaster type, environment, case study availability, and system capabilities.
- 2) It provides an overview of the use of technology in detecting humans in SAR operations.
- 3) It presents a comprehensive and in-depth review and discussion on the ML approaches used in the literature to perform human detection from UAV images in SAR missions. It also provides an overview of the main approaches used in the literature and compares them in terms of the type of images used, training dataset, model details, processing hardware, and the evaluation approach and dataset.
- 4) It reviews the available datasets on the topic and compares them in terms of type, quantity, application and environment, main purpose, and whether they were specifically built for SAR operations. It provides a brief overview of the available hardware technologies for real-time processing and compares state-of-the-art microprocessors.
- 5) It identifies and discusses the main challenges and obstacles to having a fully autonomous UAV system for detecting humans in SAR missions. Additionally, it includes recommendations on how to advance in the field and provides future directions for the research community.

III. SEARCH AND RESCUE (SAR)

Disasters are still happening and are becoming more severe and frequent over time. In recent decades, the role of SAR teams' efforts in the aftermath of a natural or man-made disaster has significantly increased [34]. Operational conditions, such as disaster sites and mountainous terrains, make SAR efforts challenging. Time is crucial, and any delay could have significant effects, including potential loss of human life [35]. To manage SAR missions in disaster situations, numerous attempts have been made. Traditional SAR methods have employed boat and helicopter approaches, which are more expensive and time-consuming. This section presents an overview of the technologies used for SAR missions over time, the different application areas, and the real case studies.

A. SEARCH AND RESCUE RESEARCH OVER TIME

The use of artificial intelligence in SAR operations began in the late 1990s with the utilization of robots and sensors. Reference [36] presented KNOBSAR as the first intelligent system prototype to offer guidance on search asset allocation of entrance points inside a disaster area. References [36] and [37] highlighted important concerns in the use of robotic

systems for SAR operations, as well as the continuous creation of a knowledge-based system for effective control of automated search equipment. Later, [38] discussed the use of serpentine robots, which are more flexible and accessible in harsh environments. Reference [39] shared their expertise of SAR missions based on their experience in training rescue teams in the fire departments in Florida. The study also discussed some issues related to using robots in SAR operations. Figure 4 shows the timeline of using artificial intelligence and technology in SAR missions.

In the early 2000s, much research was conducted to enhance the use of robots in SAR missions such as [40], [41], [42], [43], [44], and [45]. In the mid-2000s, the research communities and industry continued to enhance and integrate robots and sensors, as well as present new technologies. For example, [46] utilized RFID technology, [47] proposed an ad-hoc wireless networks to connect robots and sensors, [48] presented a distributed robot-sensor teams' approach, [49] proposed a multi robot systems, and [50] utilized an active scope camera.

The use of UAVs in SAR operations started in the mid-2000s. Reference [51] utilized mini-UAVs to capture images to assist the human-robot SAR teams. Reference [52] presented a UAV with a camera system for the purpose of scanning disaster areas, identifying victims, and providing them with some medical supplies. The system utilized a ground computer for image processing, where the system used thermal and color images. To locate shadows of the size of a person's body, a thermal image was first examined. A human body classifier was then used to classify corresponding areas in a color image, and it was set up to enable weak classifications. Reference [53] utilized machine vision to help UAVs enhance their probability of identifying missing people in maritime SAR operations. In the late 2000s, researchers tried to improve the applicability of UAVs and machine vision algorithms for SAR operations, including but not limited to [35], [54], [55], [56], [57], [58], [59], [60], [61], and [62]. There are several good resources that summarize the research done on the use of UAVs and sensors in SAR missions before 2016. For instance, [63] reviewed the literature of sensors and tracking methods used in SAR missions, [64] surveyed the literature on maritime SAR using UAVs, and [65] reviewed the research that utilized UAV with camera for detecting victims in SAR missions.

In recent years, researchers have started employing different emerging technologies in SAR operations. For instance, [9] proposed "DronAID," a real-time drone system that detects humans during disasters using PIR. Reference [66] established a system for identifying and rescuing humans from UAV images using PIR and ultrasonic sensors. Reference [67] clearly explained the purpose and working system of the DL-aided wireless sensing systems (WSSs) for human detection. Also, the authors studied the most powerful DL models that might be useful for WSSs. Moreover, the authors discussed the conventional ML and DL methods for detecting people using radar, RGB/depth cameras, and other

technologies. Reference [68] used a simulator to investigate the potential usage of motorized vehicles, drone-snowmobile Technique (DST), for SAR teams as well as drones in a mountainous terrain. The results showed that a broader area may be scanned using DST more quickly than with the traditional method, and rescuers can arrive to the victim earlier and quicker if they are delivered by snowmobile. Reference [69] proposed a cutting-edge sensing method for real-time training employing image-assisted routing, which necessitates the precise identification of people wearing sensors from UAV photos. The study suggested an integrated utilization of an accurate OD method that can outperform DL for sports situations and is built with trained filters utilizing only color features.

Reference [3] demonstrated a method that can be integrated into drones to improve the efficacy of search and rescue missions. The technique employs a drone's downward-facing camera to identify individuals and assess whether a possible landing place is safe or not. Reference [70] focused on the utilization of aerial drones to construct a human detection and geolocation system as a support for SAR missions. By calculating frame accuracy, true positive rates, and false positive rates, the study characterized the human detecting system for thermal and optical imaging. Reference [71] suggested an approach that uses nadir aerial photos and integrates DL and photogrammetric methods to identify humans on an overview orthomosaic and 3D terrain map with their coordinates. However, the model accuracy was not high. Reference [11] proposed an innovative Internet-of-Drones (IoD) system with blockchain technology, edge servers, and UAVs to enhance SAR missions. This system supports different human detection approaches while ensuring traceability, transparency, and security. Reference [72] examined techniques for using image processing to identify a previously undiscovered area. To locate people, the picture map was combined with CV algorithms. The results shown that indoor localization using QR codes is possible without the need for GPS.

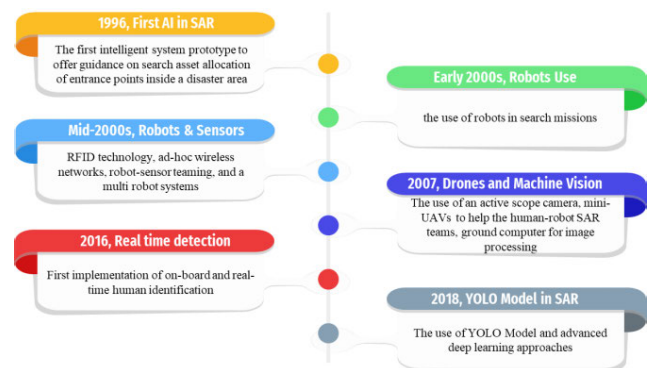


FIGURE 4. Timeline of using artificial intelligence and technology in SAR missions.

B. SEARCH AND RESCUE APPLICATIONS

This section categorizes the reviewed papers according to disaster type, area, and environment. After analyzing the

abstracts, content, and datasets used, the articles were classified into four primary groups based on the area and type of disaster. Some studies may have fit into several areas. In these instances, the authors classified those articles into the most appropriate category. According to the reviewed literature, there are five core functionalities that the reviewed systems can perform autonomously in order to enhance the efficiency and effectiveness of SAR operations:

- Object detection (OD): This task is performed when the system employs an automatic or AI method to detect objects from the aerial image.
- Human Classification (HC): This task is performed when the system uses an automatic or AI method to classify the detected object as a human.
- Autonomous navigation (AN): This task is performed when the system identifies indicators of human existence in real-time aerial videos and navigates autonomously to the probable site to acquire a clearer view and confirm the existence of people [5].
- Geo localization (GL): This task is performed when the system can automatically provide the location of the detected object.
- Thermal imaging (TI): This task is performed when the system uses thermal imaging and sensors to detect humans.

1) WILDERNESS (WiSAR)

Wilderness Search and Rescue (WiSAR) refers to the professional efforts undertaken to locate and assist individuals who are lost, injured, or stranded in isolated wilderness areas. These areas can include dense forests, mountains, deserts, coastal cliffs, and other challenging terrains. WiSAR missions are typically complex due to the restricted access, expansive search areas, and difficult navigation associated with these environments. In such extreme situations, traditional SAR approaches may prove inadequate, necessitating the deployment of specialized teams, tools, and procedures to ensure successful outcomes. Table 3 provides a comparison of the literature on WiSAR, detailing whether the work implemented the hardware system in reality, the tasks that the system is capable of, and the main contributions of the work.

WiSAR is the most extensively studied category in the literature, with more than half (50 out of 99 articles) of the reviewed literature focusing on wilderness SAR. We believe that the reasons for this include the availability of datasets for training and fewer regulations on flying UAVs in such areas. Additionally, it is not challenging to mimic or simulate a similar environment for training the ML model or validating it. Approximately 70% of the articles did not implement the hardware or test the actual system. Instead, they focused more on the software aspect, training state-of-the-art or new approaches on available or newly created datasets.

Numerous researchers have concentrated on improving and implementing the hardware component of the system, such as in [73] who presented a cutting-edge AI-based intelligent system capable of detecting people using UAVs

and multi-legged robots. Both the drone system and the multi-legged robot designs use advanced AI techniques based on CNNs. The UAV scans the area for initial detections, and the robot walks in the area to confirm the identified humans using voice detectors. In a newer study, [74] utilized UAVs, UGVs, and an edge server to complete the SAR operation. In addition, they proposed a data fusion strategy to enhance the accuracy of target identification by using RGB images, thermal infrared images, human voices, and human movements. Reference [75] developed an interactive interface that connects commercial UAVs and Head-Mounted Display (HMD) devices as mixed reality environment. It was built to handle issues such as poor identification in particular settings and the requirement for real-time video streaming. Reference [76] presented a multi-UAV cooperative platform to reduce the identification time and enhance the detection accuracy. Reference [77] evaluated factors and hardware tools (such as GPS module) that could potentially shorten the time it takes the rescue team to arrive at the target.

Some research, [76], [77], [78], [79], implemented the idea of autonomous navigation of the UAV which might be necessary in some situations such as when the system wants to take images from closer distance for better accuracy, or when the system wants to scan areas with higher probability of finding targets, or when it is beneficial in case of delivering immediate aid to the detected target. Some studies utilized GPS modules within the system to geo-locate the targets such as in [65], [73], [76], [77], [79], [80], [81], [82], [83], [84], and [85]. Some other studies used different approaches to geo-localize humans such as [86] who implemented Raycasting method in a simulated environment for person geo-localization, and [71] who suggested a new approach that uses Nadir aerial photographs and DL to identify and locate humans with GPS coordinates on a visual Orthomosaic and 3D landscape map. Only three articles implemented both autonomous navigation and geo-localization at the same time, namely [76], [79], [80]. Finally, only [79] who implemented the five tasks all together.

2) MARINE SAR (MSAR)

Marine Search and Rescue (MSAR) is dedicated to locating and rescuing individuals who have gone missing or are in danger in maritime environments. Unlike WiSAR and USAR, MSAR operations necessitate an immediate response to rescue individuals. Table 4 provides a comparison of the literature on MSAR, indicating whether the work implemented the hardware system in reality, the tasks that the system can perform, and the main contributions of the work.

MSAR has been extensively studied in the literature, with 16 of the reviewed articles focusing on marine SAR. Similar to WiSAR, there are available datasets for training, fewer regulations on using UAVs in such environments, and the possibility to simulate similar scenarios for data collection and system validation. Approximately 62% of the papers did not implement or test the actual system. Instead, they focused more on the software aspect, training cutting-edge,

TABLE 3. Summary on the WISAR literature.

Article	Case Study	OD	HC	AN	GL	TI	Main Contribution
[65]	Yes	X			X		Presented a new approach based on color signature algorithm and implemented a case study
[81]	Yes	X			X		Presented a new approach and implemented field experiments on forests, mountains, and wilderness areas, particularly under time restrictions and harsh weather conditions.
[87]	No	X	X				Presented a new approach and trained it on Avalanche environments
[79]	Yes	X	X	X	X	X	Integrated YOLO model with airborne optical sectioning to analyze thermal images in forests environments.
[73]	Yes	X	X		X	X	Main contribution is on the hardware by integrating UAVs with multi-legged robots.
[88]	No	X					Proposed a new approach and trained it on a new dataset of plain, forest, and sea environments
[86]	No	X	X		X		Implemented Raycasting method in a simulated environment for person geo-localization for the task of finding missing people.
[89]	No	X	X				Trained several models on available datasets and tested their performance on a new dataset.
[82]	No	X	X		X		Integrated CNN with SVM to identify humans in Avalanche environments and compared the results with different detection models.
[90]	Yes	X	X				Presented a new approach
[91]	No	X	X				Tested different models on available datasets
[92]	No	X	X				Presented a new approach that combines ensemble learning model with EfficientDET architecture for mountain and landscapes environments
[93]	No	X	X				EfficientDET deep NN is trained using a newly generated dataset
[71]	No	X	X		X		suggested approach uses Nadir aerial photographs and integrated DL and photogrammetric techniques to identify and locate humans with GPS coordinates on a visual Orthomosaic and 3D landscape map.
[94]	No	X	X				Presented a new dataset with different images' heights and tested different state-of-the-art algorithms
[95]	in use	X	X				Presented a new approach and trained it on a new dataset.
[96]	No	X	X				Enhance tiny objects detection that could be covered or obstructed in the images.
[97]	Yes	X	X				Presented a case study of a real implementation for detecting and rescuing a human using UAV and CNN
[98]	No	X	X			X	The research studies the effect of environmental conditions on the accuracy of the state-of-the-art algorithms
[99]	No	X	X				Implemented and studied the effectiveness of YOLOv3 Model and compared its results with F-RCNN, SSD, and R-FCN
[3]	No	X	X				Presented a new dataset and proposed a new approach.
[75]	Yes	X	X				It used Head-Mounted Display (HMD) devices that provide Mixed Reality to Ground Stations. Overcoming issues such as poor identification in specific settings.
[100]	Yes	X	X				Presented a new dataset, tested five versions of YOLOv5 model, and conducted a simulated SAR experiment to test the system.
[101]	No	X	X				Trained the lightweight YOLO model in two datasets and compared the results with the literature.
[80]	Yes	X	X		X		Presented a real implementation of the hardware and software, and compared between different detection algorithms.
[102]	No	X	X				New approach employs sequences of sequential pictures rather than just one static picture.
[103]	No	X					Compared the time and efficiency over time between manual and computer assisted approaches within a simulated SAR operation.
[104]	No	X	X				Trained YOLO on SARD and compared it with other detectors
[105]	No	X	X			X	Trained YOLO model on thermal imaging to mitigate the impact of bad weather.
[106]	No	X	X				Presented a new dataset and tested different state-of-the-art algorithms
[107]	No	X	X				Presented a new multimodal method and used contextual data in the surrounding area to enhance the identification accuracy in nonurban terrain.
[76]	Yes	X		X	X		Presented a multi-UAV cooperative platform to reduce the identification time and enhance the detection accuracy
[108]	No	X	X				Presented a new approach to enhance the accuracy of detection
[109]	No	X	X				Presented a new dataset, trained the Faster R-CNN, and tested the performance with other methods
[110]	No	X	X				Collected new dataset on mountains and beaches areas and trained tiny YOLO model.
[83]	No	X	X		X		Presented a new dataset and trained YOLO model
[111]	No	X	X				Trained the SSD model and tested it.
[112]	No	X	X				Proposed a new ML algorithm by extending YOLO model and tested it on a new dataset.
[113]	No	X	X			X	Trained two ML algorithm, YOLO, and Faster R-CNN, and tested the performance on a new data
[78]	No	X		X			Presented a system for detecting objects using anomaly detection algorithm and tested it on a new dataset
[114]	Yes	X	X				Proposed a robust system design with a new CNN-based algorithm connected to smartphone for real-time detection
[84]	Yes	X	X		X		Implemented the system, used YOLO model, and did some flight tests
[115]	No	X	X			X	Implemented a semi-autonomous system that uses thermal and RGB cameras with a YOLO model trained on a new dataset.
[77]	Yes	X	X	X			Focused on improving the hardware implementation and assessing the factors that reduce the time it takes the rescue team to get at the location.

TABLE 3. (Continued.) Summary on the WISAR literature.

[116]	No	X	X			X	Presents a novel approach for human detection and tracking, collected a new dataset, and tested out the approach on three videos
[85]	No	X	X		X		Trained the YOLO model on a new dataset and implemented the system.
[117]	No	X	X				Finetuned the YOLO model on three datasets of urban, wilderness, mountains environments
[74]	Yes	X	X			X	Presented a system of UAVs and AGVs with YOLO and data fusion method to detect humans from RGB images, thermal images, and human voice.
[118]	No	X	X			X	Collected a thermal dataset on beach and forest environments, trained YOLO model and validate the model with flight tests
[119]	Yes	X	X				Presented a new DCNN algorithm and trained it on HERIDAL and new collected images. The performance of the algorithm was compared to human experts' image analysis.

or novel techniques on existing or new datasets. A few of these studies tested their systems using computer simulations, such as [120], [121], and [122]. For instance, [120] built a high-fidelity dynamic simulation model for a marine environment to optimize the setup of UAV and human detection models to obtain the highest possible accuracy. On the other hand, the main contributions of the articles that did implement the hardware system were in the software part too. However, [123] examined vision-based navigation and management of a joint unmanned surface vehicle (USV) and UAV system for MSAR.

Some studies utilized GPS/GNSS modules within the system to geo-locate detected humans such as [10], [124], [125], and [126]. Three articles in MSAR had implemented AN on the UAV [10], [122], [123] and only two articles in MSAR performed both AN and geo-localization simultaneously [10], [122]. In addition, one of the articles, [122], did implement the five tasks all together.

Among the reviewed literature, only [52] implemented aid delivery within the proposed system. They discussed an emergency services operation scenarios that used UAVs to identify and rescue injured humans. The system involves UAVs to scan selected regions and attempt to locate injured people and then attempts to provide medical and other assistance to the designated individuals.

3) URBAN SAR (USAR)

Urban Search and Rescue (USAR) refers to technical rescue operations in urban settings where individuals have become trapped, missing, or stranded due to disasters such as earthquakes, storms, and fires. These disasters can result in full or partial building collapses, posing significant challenges to rescue efforts. Unlike Wilderness SAR, which deals with remote wilderness areas, USAR operations occur in densely populated urban areas. USAR teams face challenges such as navigating through debris, dealing with structural hazards, and covering wide search areas. Table 5 compares the literature on urban SAR, indicating whether the work implemented the hardware system in reality, the tasks that the system can perform, and the main contributions of the work.

USAR has been less studied in the literature, with only 9% of the articles focusing on urban SAR. Unlike the other two environments, there are fewer available datasets for training, more regulations on flying UAVs in such areas, and it is

difficult to construct a simulated environment. More than half of the reviewed literature on USAR implemented the hardware and tested the actual system. These studies focused more on the hardware aspect, training state-of-the-art or new approaches on available or new datasets. The studies that concentrated on improving and implementing the hardware part of the system are [5], [52], [54], [135], [136], [137], and these were the only five studies utilizing GPS modules within their system to geo-locate the targets. Only one study, [5], implemented AN on their UAV system and geo-localization at the same time. However, none of the articles implemented the five tasks together.

Most of the articles utilized both RGB camera and thermal imaging to detect targets including [52], [54], [137], [138], [139], [140].

4) OTHER ENVIRONMENTS

Other articles deal with or are conducted within different environments, such as indoor environments, actual disasters, and volcanic areas. Table 6 lists and compares the literature on these other environments. The table indicates whether the work implemented the hardware system in reality, the tasks that the system is capable of, and the main contribution of the work.

IV. MACHINE LEARNING MODELS

This section discusses the various ML approaches used in the reviewed literature and compares them in terms of training dataset, type of input, processing hardware, and evaluation method and datasets. We will discuss the three main models used to detect humans from aerial images, including YOLO, Fast RCNN, and Single Shot Detector (SSD). Subsequently, we will discuss other approaches proposed by the reviewed articles and conduct a comparison among these approaches.

Around 2010, as the accuracy of individually created features got saturated, OD reached a saturation point. However, convolutional neural networks (CNN) made their debut in 2012, when deep CNNs demonstrated their effectiveness in learning reliable and sophisticated visual representations of an image. The introduction of Regions with CNN features (RCNN) for object detection in 2014 resolved the stalemate in object detection [151].

Due to its ability to solve a wide range of difficult problems, DL has recently taken the lead in the CV sector.

TABLE 4. Summary on the marine SAR literature.

Article	Case Study	OD	HC	AN	GL	TI	Main Contribution
[126]	Yes	X	X		X		Trained a new approach on two new datasets and conducted a case study
[120]	Simulation	X	X				Used High-Fidelity Dynamic Marine Simulation Environment to train and test YOLO model
[127]	No	X	X				Fine-tuned the YOLO model
[128]	Yes	X	X				Presented a new dataset and conducted a comparison between different models
[129]	No	X	X		X		Adding modules (AFPN, BiFormer) to the YOLO to enhance the performance
[130]	Yes	X	X				Trained a new approach on a new dataset and conducted a case study
[121]	Yes	X	X				Presented a small new dataset, and YOLO finetuning
[123]	Simulation	X	X	X			real-time UAV navigation and control
[131]	No	X					Introduced a new adaptive image processing system to handle constantly changing backgrounds by analyzing color and identify frequency patterns using a vision camera (EO camera).
[132]	No	X	X				Collected a new dataset and trained an online commercial ML model and tested it.
[133]	No	X	X				Trained and tested YOLO on a new dataset of photos with various altitudes.
[122]	No	X	X	X	X	X	Trained the R-CNN on a new thermal dataset and proposed an approach for locating the target.
[125]	Yes	X	X		X		Used a new dataset to train and test YOLO model. It presents an approach to locate the target and send the location to the SAR team
[134]	No	X	X				Modified the YOLO model and integrated it with the Simple, Parameter-Free Attention Module (SimAM) to identify focus regions in the image.
[10]	No	X	X	X	X		Presented a new dataset and trained YOLO model on a fully autonomous system
[124]	Yes	X	X		X		Presented a new dataset and trained both Faster R-CNN and SSD-Resnet model for human detection, geolocating, and tracking on a fully autonomous UAV

TABLE 5. Summary on the urban SAR literature.

Article	Case Study	OD	HC	AN	GL	TI	Main Contribution
[52]	Yes	X			X	X	Presented a new approach and conducted a case study on asphalt, roads, grass, building roof, trees, and water environments
[5]	Yes	X	X	X	X		New approach (AN) and conducted a simulated case study on grass and asphalt environments.
[141]	No	X	X				Presented a new approach and trained it on a dataset collected on grass and asphalt environments
[135]	Yes	X	X				The main contribution is on implementing the hardware and presenting a new ML approach
[54]	Yes	X	X		X	X	New image processing framework for detecting humans from thermal and color images on asphalt and gravel roads
[136]	Yes	X	X		X		Proposed a new approach and tested the system in real flights
[138]	No	X	X			X	Presented a new dataset (campus, grass, and seashore environments) and trained YOLO model
[139]	No	X	X			X	Presented a new dataset (campus, grass, and seashore environments) and trained YOLO model
[140]	No	X	X			X	Presented a new dataset (campus, grass, and seashore environments) and trained YOLO model
[137]	Yes	X	X		X	X	Presented a new system architecture and trained YOLO on RGB and infrared images

Various DL designs, including CNNs, recurrent neural networks (RNN), and generative adversarial neural networks (GAN), are employed for CV, particularly OD [89]. The CNNs are perhaps the most popular models among them. Deep CNNs, like YOLO [152], Faster-RCNN [153], RetinaNet [154], Cascade R-CNN [155], and SSD [156], have lately made it possible for machines to identify things in photos even more accurately than humans.

Reference [30] thoroughly examined the applications and uses of DL for UAVs, considering the most important developments as well as their strengths and weaknesses. The authors concluded by describing the main challenges in using DL for UAV-based applications. Reference [67] clearly explained the purpose and working system of the DL-aided wireless sensing systems (WSSs) for human detection. In addition, the authors studied the most powerful DL models that might be useful for WSSs. Moreover, the authors discussed the conventional ML and DL methods for detecting people using radar, RGB/depth cameras,

and other technologies. Reference [119] established several image-processing techniques to facilitate and accelerate mission image processing and improve detection accuracy. The most recent ones relied on CNNs which was trained on specially created image database called HERIDAL. The authors developed a web application to allow professionals to assess unprocessed mission images. When analyzing images that had already been altered and labeled, experts were able to interpret them with greater accuracy than the algorithm.

A. YOU ONLY LOOK ONCE (YOLO)

The You Only Look Once (YOLO) algorithm, developed by [152], represents a cutting-edge, open-source, real-time object detector that has garnered significant attention in the field of CV. YOLO offers several versions, with YOLO v3 being the most widely used, featuring a Darknet-53 architecture comprising 106 fully convolutional layers. YOLOv10 was the most recent version made available.

TABLE 6. Summary of the literature on the other SAR environments.

Article	Application	Case Study	OD	HC	AN	GL	TI	Main Contribution
[142]	Indoor	indoor	X	X		X		To test different algorithms including HOG, YOLO v3, and Mask RCNN on dataset of indoor scenarios. It also, tested the localization of the humans in the environment.
[143]	Indoor	indoor	X	X				Presented a new indoor dataset and implemented the system and tested it indoor
[144]	Indoor	indoor	X	X				Presented a new dataset of human gestures, trained YOLO model and implemented and tested the system in the lab
[145]	Indoor	indoor	X	X			X	Used pretrained YOLO model on a thermal dataset and tested it out
[146]	Indoor	indoor	X	X	X			Presented a system for real-time automatic detection and navigation and validate it with flight tests.
[147]	Actual disasters	No	X	X				New CNN model was trained on three small datasets from actual disasters. And a simulation system was implemented for testing the approach.
[148]	Volcanic areas	No	X	X				Test, finetune, and optimize the MobileNetV2 SSD model. But the dataset was not specific to SAR
[149]	Various scenes	Yes	X	X				The use of 3D-LiDAR sensor and new CNN approach for SAR
[150]	Various scenes	No	X				X	Trained the YOLO model to detect objects from infrared images.

Unlike traditional object detection methods that relied on adapting classifiers for detection tasks, YOLO suggests predicting boundary squares and probability of classes concurrently with a single neural network. This end-to-end method simplifies and automates the OD task, resulting in cutting-edge performance that surpasses other real-time OD models significantly. YOLO uses a single fully connected layer to achieve all of its predictions. While techniques utilizing region proposal networks execute numerous iterations on a single image, YOLO accomplishes the task in a solitary iteration, contributing to its efficiency [29], [152], [157].

YOLO algorithm begins by receiving photos as input and utilizes a straightforward deep CNN to identify targets within the photo. YOLO divides photo into an $S \times S$ grid, where an object's center-containing grid squares are in the position of detecting it. Within every cell, YOLO draws B bounding squares along with confidence levels, showing the precision of the square's estimate and the model's certainty of the existence of an object inside the box.

Central to YOLO's approach is its strategy of predicting several bounding boxes within each grid cell. Among these, one predictor is designated as "responsible" for a particular target determined by the ground truth's highest Intersection over Union (IOU). This fosters specialty among predictors, enhancing the overall recall score of the model. Furthermore, YOLO models employ non-maximum suppression (NMS) as a crucial technique to improve object detection accuracy and efficiency. Acting as a post-processing step, NMS addresses the issue of multi bounding squares being created for one target in each photo by identifying and eliminating incorrect or irrelevant bounding boxes leaving only one bounding square for each target in the photo. The YOLO algorithm, with its innovative approach to object detection, has made significant strides in the field of CV, offering real-time

capabilities and superior performance compared to traditional methods.

The process begins by resizing the input image to dimensions of 448×448 before passing it to the CNN. Initially, a 1×1 convolutional layer is utilized to reduce the channels number, then a 3×3 convolutional layer to generate a cuboidal output. Rectified Linear Unit (ReLU) is utilized as an activation function throughout, except for the last layer, which uses a linear function. Figure 5 shows the architecture of the YOLO model.

YOLO is the most widely used framework for identifying humans in photos and videos from UAVs in SAR missions, with approximately half of the articles in this field utilizing it. Table 7 provides a comparison of the literature that has employed YOLO in their studies. The table includes the type of input images, the training dataset, the model version, the processing hardware, and the evaluation and validation dataset.

Some articles tried to improve the YOLO model by adding new features and capabilities. For instance, [96], [129] presented a modified YOLO model by incorporating feature pyramid network (FPN) to maintain the target feature information. Reference [129] also improved the model's ability to perceive small-scale objects by a new BiFormer component and solved issues between localization and classification by the task-specific context decoupling (TSCODE) technique. Reference [112] added various enhancements to the YOLO Model, including "finer strides" and a new "third output layer" to enhance low-pixel person identification on thermal pictures. Reference [134] included a prediction head to identify small-scale individuals or targets. Furthermore, they used the Simple, Parameter-Free Attention Module (SimAM) to identify interest points in the picture.

Other research trained and tuned the YOLO model to detect humans from both types of input images, RGB and Thermal. For example, [73] presented an edge AI-based robot platform

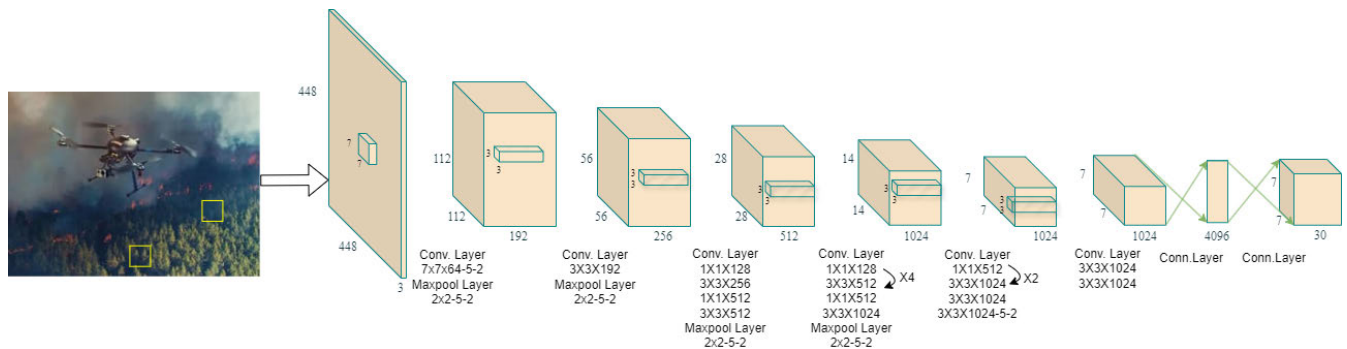


FIGURE 5. YOLO model architecture.

with UAVs and multi-legged robots for SAR purposes. The platform uses Tiny-YOLO on the UAVs to analyze RGB and thermal infrared images and it uses a resource-efficient quantized NN on the robot to analyze human voice and activities to accurately identify humans. In another study, [74] employed YOLO to identify individuals in real time, where an image fusion approach that combines RGB and thermal infrared images was suggested to improve human identification accuracy. In addition, a data fusion strategy was proposed to enhance the comprehensiveness of the detection, which incorporates RGB images, thermal infrared images, human sounds, and human activities. Reference [98] evaluated YOLOv3 and CNN-PDI performance on 8 different datasets. YOLOv3 achieved the best average F-1 value and precision. Then, they observed that YOLO's RGB edition struggled to recognize objects labeled 'person'; instead, the object was identified as 'bird'. Nevertheless, the infrared edition reduced the YOLO's false positive scores to zero throughout straight and orbital flight movements. Reference [115] utilized off-the-shelf RGB and thermal cameras and claimed that the performance is similar to the literature.

Some research did real field experiments to validate and test the system and model performance such as [73], [79], [84], and [118]. Other research did computer simulation studies to validate the system and model such as [86] and [120].

Much research has been done for real-time human detection from aerial images. For instance, [10] offered a real-time person detection technology that may be used on board to find open water swimmers in emergency marine scenarios. The system processes the photos that the drones capture using the YOLO DL model. The outcomes demonstrated that the model is capable of achieving a high level of detection accuracy in marine emergencies. But the model is inappropriate for other kinds of disasters. Reference [10] also suggested a fully autonomous UAV equipped with a real-time people detecting system. The system used the YOLOv3 DL approach to detect swimmers in open water using the Nvidia Jetson TX1 embedded hardware. The novel aspect of the proposed system is its integration of Global Navigation Satellite System (GNSS) techniques with CV techniques to enable precise human identification and SAR operation implementation. Reference [83] presented Project Vulture, a potent method for localizing

human subjects that speeds up the processing of collected drone aerial pictures for dispersed and scalable operations. The framework was constructed using a trained DL model, YOLO v3.

Later, [144] studied the real-time UAV human identification of body and hand rescue hand signs. To create biometric communications, the authors leveraged body-featuring technologies, such as yolo3-tiny for human detection. Using the YOLOv3, the system achieved excellent accuracy of about 99.80 % on testing data for body gestures and 94.71 % on testing data for hand gestures. Reference Reference [114] used a unique system that combined a UAV with a real-time ML-based object recognition system (YOLOv3) embedded on a smartphone to carry out the SAR missions. The findings showed that the system provided a very good accuracy of 94.73 % and a frame rate of 6.8.

Recently, [143] used YOLOv2 model based on the drone's camera for the creation of systems for the identification and detection of people. DL-based CV algorithm was used to detect the person's position and state. The findings of the person detection demonstrated a significant degree of object detection and classification accuracy using YOLO-v2. The tracking algorithm reacted quicker for real-time tracking than commonly used methods, effectively tracking the discovered human without missing it from sight. Reference [157] examined the YOLO model's detection process and functionality on three different accelerator-based single-board computers (SBCs): the Raspberry Pi 4B (RPi) with Intel Neural Compute Stick2 (NCS2), the NVIDIA Jetson Nano, and the NVIDIA Jetson Xavier NX (NCS2). The RPi+NCS2's capabilities are more accommodating to lightweight models. Moreover, the procedure of using Jetson devices is significantly simpler than running a model on RPi + NCS2. The Jetson Nano is an SBC that compromises performance and price.

B. FAST REGION-BASED CNN (FAST RCNN)

The Fast R-CNN framework represents a major milestone in the area of OD, offering a streamlined and efficient approach to detecting objects within images. Developed by [160], Fast R-CNN integrates proposal generation and feature extraction into a single, end-to-end framework, thereby alleviating

TABLE 7. Summary of the literature that used YOLO models.

Article	Images type	Training Dataset	Model	Processing	Evaluation
[120]	RGB	SeaDronesSee	YOLOv7	Real-time GS	Simulation
[127]	RGB	COCO	YOLOv3	GS	Same dataset
[79]	Thermal	Thermal data, Darknet	YOLOv4 & adaptive online sampling for path planning	Onboard	Field experiments
[73]	RGB and Thermal	New	Tiny-YOLO	Onboard	Field experiments
[129]	RGB	MOBDrone, SeaDroneSee	ABT-YOLOv7	GS	Same Dataset
[86]	RGB	New dataset	YOLOv4	GS	Simulation
[121]	RGB	Three datasets: one from a video on YouTube and two new	YOLOv4-tiny	Onboard	Two new datasets
[91]	RGB	VisDrone2018, [158]	YOLOv3: YOLOv3-SPP1, SlimYOLOv3, Mask Faster R-CNN	GS	Same dataset, models' results comparison
[94]	RGB	New dataset	ACF, YOLO MobileNet, and YOLO ResNet50	GS	Same dataset
[96]	RGB	SARD	YOLO and modified Feature Pyramid Network	GS	SARD
[98]	RGB and Thermal	New dataset	YOLOv3, CNN-PDI	GS	8 different datasets
[99]	RGB	DOTA	YOLOv3	GS	DOTA
[143]	RGB	new dataset includes images taken from both long and close distances	YOLO v2	Onboard	Same dataset
[100]	RGB	New dataset with various flight patterns and magnification ranges	Five different versions of YOLOv5	Onboard	Same dataset
[101]	RGB	HERIDAL and SARD	YOLOv5	Onboard	Same dataset
[104]	RGB	SARD	YOLOv5L	Both	Same dataset
[105]	Thermal	PD T dataset [159]	YOLOv4	GS	Same dataset
[76]	RGB	VisDrone2019, and new images	YOLOv5	GS	Same datasets
[150]	Thermal	HIT-UAV	YOLOv5	GS	Same datasets and COCO
[110]	RGB	Collected new dataset from YouTube videos	Tiny-YOLOv3	Onboard	Same datasets
[83]	RGB	New dataset	YOLO v3	GS	Same dataset
[144]	RGB	new	YOLOv3	Onboard	Same dataset
[112]	Thermal	new	New algorithm extending YOLO	Onboard	Same dataset
[138]	Thermal	New dataset	YOLOv3	Onboard	Same dataset
[139]	Thermal	New dataset	YOLOv3, YOLOv3-MobileNetV1 and YOLOv3-MobileNetV3	Onboard	Same dataset
[114]	Optical	New dataset for diverse outdoor environments	YOLOv3	Onboard	Same dataset
[84]	RGB	Used the pretrained YOLO	YOLOv3	Onboard	Flight tests
[115]	RGB and thermal	Images of actual soccer videos on YouTube	YOLOv5	GS	Same dataset
[116]	Thermal	New dataset	YOLOv5	GS	Same dataset
[85]	RGB	New dataset	YOLOv3	Onboard	Same dataset
[133]	RGB	New dataset	YOLOv3	Onboard	Same dataset
[117]	RGB	SARD, VisDrone, Corr-SARD (added weather effects)	YOLOv4	GS	Same datasets
[74]	RGB and thermal	Pretrained model	YOLO	GS	New images
[140]	Thermal	New dataset	YOLOv3, YOLOv3-MobileNetV1 and YOLOv3-MobileNetV3	Onboard	Same dataset
[137]	Thermal	FLIR thermal	YOLOv3, YOLOv4, YOLOv7	Onboard	Same datasets
[118]	Thermal	New dataset (OTCBVS)	YOLOv3	GS	test flight
[145]	Thermal	COCO	YOLOv5	GS	Same datasets
[10]	RGB	New dataset	Tiny YOLOv3	Onboard	Same dataset
[125]	RGB	New dataset for close range aerial HD	YOLOv5	Onboard	Same dataset
[134]	RGB	SeaDronessee	YOLOv7 and SimAM	GS	ODv2 challenge dataset

the computational burden associated with traditional object detection pipelines [153], [160].

While YOLO achieves all predictions through a single fully connected layer, Faster R-CNN employs a two-step process involving region proposal and individual recognition of these regions. At the heart of Fast R-CNN lies the concept of region-based CNN (R-CNNs), which enable the model to focus its attention on specific regions of interest (RoIs) within an image. By leveraging region-based CNNs, Fast R-CNN efficiently extracts relevant features from RoIs, reducing redundancy and computational overhead [153], [160].

The overall workflow of Fast R-CNN can be summarized as follows:

- **Input Processing:** The input image undergoes convolutional processing through a CNN to generate convolutional feature maps, serving as the basis for RoI identification.
- **Region Proposal:** Fast R-CNN employs a region proposal network (RPN) to identify potential RoIs where objects may be present. These proposed RoIs are predicted bounding boxes around potential objects.
- **RoI Pooling:** An RoI pooling layer is applied to extract features from each RoI. This layer ensures that features from RoIs of varying sizes are properly aligned and standardized, facilitating their integration into subsequent network layers.
- **Classification and Localization:** Features extracted from RoIs are fed into a fully connected network (FCN) for both classification and localization tasks. Through a combination of softmax classification and bounding box regression, Fast R-CNN predicts object presence within each RoI and refines bounding box coordinates.
- **Output:** The final output comprises predicted classes for each RoI, along with corresponding bounding box coordinates.

Fast R-CNN's efficient design and performance improvements have solidified its position as a cornerstone in CV research, paving the way for further advancements in object detection and related tasks. Figure 6 presents the architecture of the Fast R-CNN model. Fast R-CNN was quite frequently used in detecting humans from UAV images and videos in SAR missions. Table 8 lists and compares the literature that used it in their study. The table lists the type of the input images, the training dataset, model version, the processing hardware, and the evaluation and validation dataset.

Fast R-CNN was used by researchers recently in SAR operations to detect humans from aerial images. For instance, [5] provided a drone-based human identification system in order to improve the effectiveness of SAR operations. After identifying human signs in real-time aerial videos, the drone automatically navigates to the suspected area to acquire a closer look and confirm human presence. The proposed object detector, Fast RCNN, analyzes photos in real-time and reports human objects if they are found on the ground. Reference [109] examined the efficiency of human detection in drone photos using the VisDrone, Okutama-Action, and

SARD datasets which are created to represent SAR missions. The model was a Faster R-CNN with FPN as the base, pre-trained on COCO dataset. References [5] and [126] conducted flight tests to evaluate and validate the model performance. Reference [122] used the model to detect humans from thermal images. Reference [126] integrated the GPS technology into a Fast RCNN to identify, locate and map humans in flooding SAR missions. The algorithm also detects the possible obstacles to the drone to avoid them.

C. SINGLE SHOT DETECTOR (SSD)

Single Shot Detector (SSD) algorithm represents a significant advancement in the area of OD, particularly within the domains of CV and ML. Introduced by [156], SSD has gained widespread recognition for its effectiveness in identifying objects within images, notably in applications such as aerial image analysis for SAR missions. SSD's breakthrough lies in its real-time capabilities, which allow it to detect multiple objects within images with high precision. The algorithm operates by taking an image as input and simultaneously predicting bounding boxes and labels for all targets present in the photo. This unique "single shot" approach combines localization and classification tasks, enhancing its efficiency and effectiveness.

At the heart of SSD is a deep CNN architecture, specifically designed for object detection. The network architecture integrates convolutional layers with varying spatial resolutions to recognize targets of various scales and aspect ratios. Notably, the last layer is responsible for predicting both class labels and offsets for bounding boxes, facilitating accurate object recognition and localization. Owing to its high-speed and efficient design, SSD is particularly perfect for real-time tasks like autonomous driving. Its exceptional performance across various object detection benchmarks has established it as a preferred choice for numerous CV applications.

SSD algorithm, as introduced by [156], represents a significant leap forward in object detection technology. Its combination of real-time capabilities, efficiency, and accuracy makes it a valuable tool for a diverse array of CV tasks. Figure 7 presents the architecture of SSD model. Table 9 lists and compares between the literature that used SSD model on their research. The table lists the type of the input images, the training dataset, model version, the processing hardware, and the evaluation and validation dataset. For instance, [141] proposed a method for human detection and action recognition inspired by the pyramidal feature extraction of SSD. When tested on a new collected dataset, the proposed model achieved excellent accuracy. Reference [146] implemented SSD and did some flight experiments to validate the model's performance. All articles used the model to detect humans from RGB images only.

D. OTHER DEEP LEARNING ALGORITHMS

There are other ML approaches used in the literature to detect humans from aerial images. Table 10 lists and compares the literature that used these different approaches in their

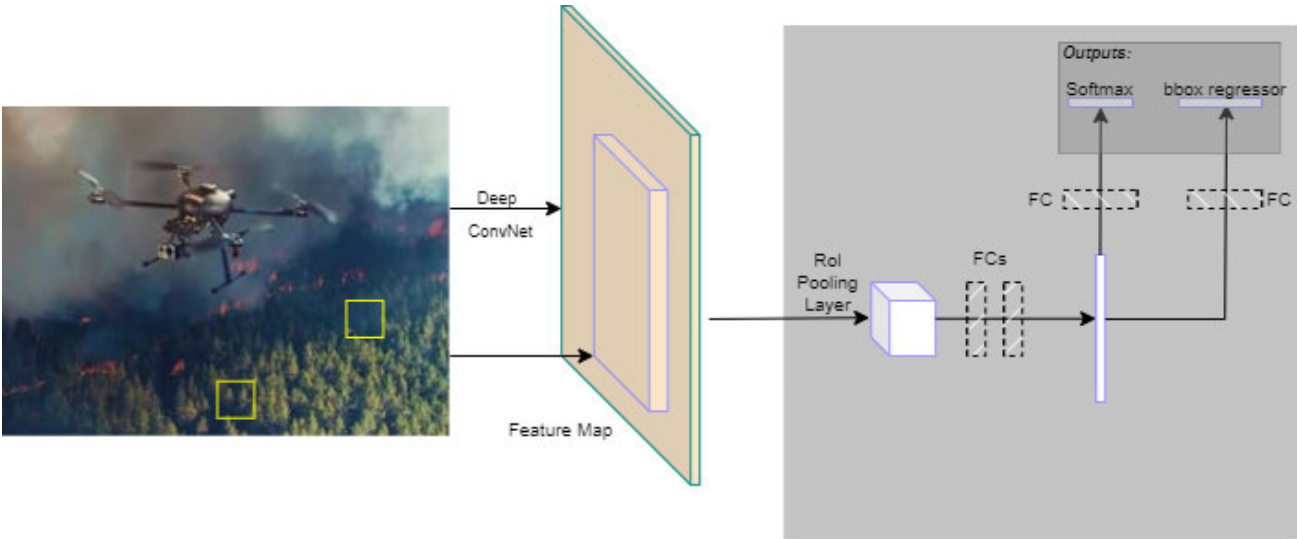


FIGURE 6. Fast R-CNN model architecture.

TABLE 8. Summary of the literature that used fast R-CNN models.

Article	Images type	Training Dataset	Model	Processing	Evaluation
[126]	RGB	Two new datasets: day and dusk	Fast R-CNN	Onboard	Other datasets and Field experiments
[109]	RGB	VisDrone, Okutama-Action, SARD	Faster R-CNN	GS	Same datasets
[122]	Thermal	New dataset	Faster R-CNN	Onboard	New dataset
[124]	RGB	New dataset	Faster R-CNN and SSD-Resnet	Onboard	Same datasets
[5]	RGB	Okutama-Action	Fast RCNN	Onboard	simulations and actual flights

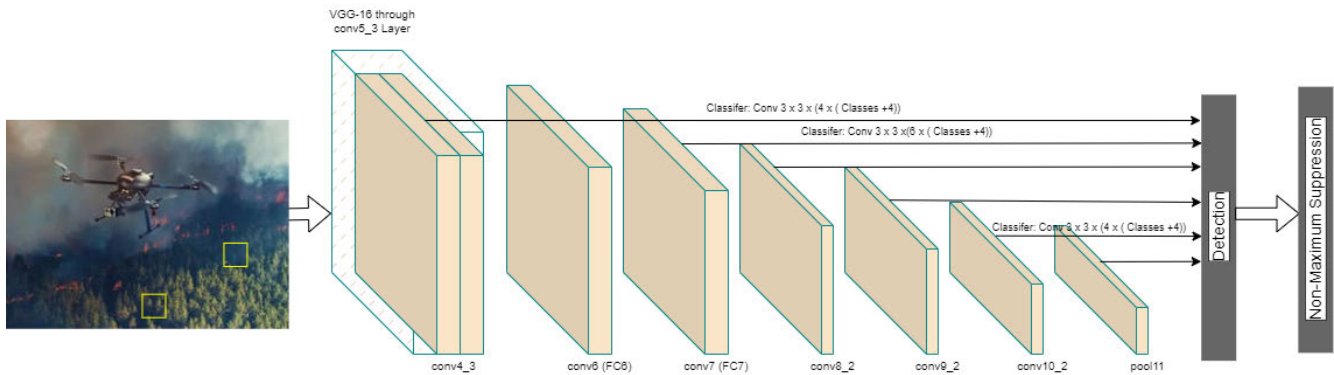


FIGURE 7. SSD model architecture.

research. The table lists the type of the input images, the training dataset, model version, brief about the approach, the processing hardware, and the evaluation and validation dataset. Many other detectors had been proposed and suggested by researchers to be used in the UAVs for SAR operations. For instance, [65] proposed a fully autonomous fixed-wing UAV that is created and equipped with an all-in-one camera-based object recognition and localization system. The system is a two-stage framework for identifying and locating humans in SAR operations. The first

stages used a color signature algorithm onboard to identify targets. The second stage uses the recorded video, flight logs, and GPS system to identify and locate the targets. The system was effective of post-target identification and location as well as the capture of aerial images for additional mapping applications in addition to on-board, real-time target identification. The suggested algorithm can detect humans, crushed cars, or aircrafts by using the color signature. Reference [161] examined salient object recognition methods using Synthetic Aperture Radar photos collected by

UAVs for SAR missions. It also investigated the behavior of salient objects detection algorithms for pictures taken in natural settings on relatively uniform grassy topography and diverse topography. Reference [87] proposed an integrated approach between CNN and SVM to detect humans in avalanche's SAR missions. The proposed approach has a pre-processing strategy that improves the identification accuracy and a post-processing technique that relies on a Hidden Markov Model to enhance the classifier's accuracy. Reference [81] provided an approach for identifying humans in wilderness locations and calculating their precise locations to assist ground searchers in locating them within a specified time range. The authors tested the suggested system in a real-world search situation, particularly under time restrictions and harsh weather conditions. The approach employs a ring model to determine a missing person's position by creating a ring around the Point Last Seen (PLS). It also uses a mobility model that considers terrain characteristics, creating isochrones to represent possible regions within a given time frame. Finally, the system applies a nested k-means algorithm that analyze aerial photos to determine probable positions of missing human. Reference [71] suggested an approach that uses nadir aerial photos and integrates DL and photogrammetric methods to identify humans on an overview orthomosaic and 3D terrain map with their coordinates. However, the model accuracy was not high. Reference [69] proposed a cutting-edge sensing method for real-time training employing image-assisted routing, which necessitates the precise identification of people wearing sensors from UAV photos. The study suggested an integrated utilization of an accurate OD method that can outperform DL for sports situations and is built with trained filters utilizing only color features.

In the last a few years, [93] developed a novel model for detecting people in UAV images for SAR missions. The model was based on EfficientDET deep neural network and obtained higher accuracy compared to what the Croatian Mountain SAR teams uses currently. Reference [130] presented an ensemble algorithm consisting of several deep CNNs that were orchestrated by the fusion module with non-linearly adjusted voting weights. The technique surpasses all the cutting-edge deep neural networks, including YOLOv3, -v4, Faster R-CNN, RetinaNet, and SSD300. However, the model is trained to detect objects floating on water surfaces. Reference [95] provided a new deep neural network-based human detection approach that combines dilated convolutions with Path Aggregation Network (PAN) for real time SAR. The accuracy of the proposed model is somewhat close to the standard YOLO v3 model; yet the speed is twice as fast as the typical YOLO v3. However, YOLO v4 is marginally more accurate than the proposed model.

Recently, [92] introduced a unique approach based on the EfficientDET architecture and ensemble learning to identify people in aerial images of mountainous areas taken by UAVs operating in SAR missions. The approach achieved the best detection performance on the HERIDAL dataset. Reference [102] continued and further enhanced the previous

research in detecting humans from aerial images for SAR operations in term of the detection outcomes and precision value. The authors introduced a novel method that employs sequences of succeeding images as input to the multimodel neural network architecture rather than a single static image. The same object of interest needs to be found in multiple images since succeeding images overlap.

E. COMPARISON AMONG DIFFERENT APPROACHES

In the recent years, much research has been done to compare among available DL algorithms for detecting humans from UAVs images and videos in SAR operations (Table 11 shows a summary of the literature that compare between different models).

Reference [91], analyzed various DL detectors for the purpose of detecting people from UAV images in volcanic activity SAR operations. The results showed that Mask-Fast RCNN has the highest detection accuracy, YOLOv3, tiny YOLOv3, and SSD came next. SlimYOLOv3 outperformed competitively when weighed against examples developed on the Mask Faster RCNN and YOLOv3 SPP1 networks, requiring a substantially longer period of inference time. Additionally, Slim YOLOv3 is significantly slimmer in the number of parameters. Reference [89] explored the efficiency of available cutting-edge detectors like Faster R-CNN, YOLOvs, RetinaNet, and Cascade R-CNN on VisDrone and SARD datasets. The findings illustrated that on SARD dataset, YOLOv4 had the best detection accuracy.

Reference [163] reviewed several DL algorithms for real-time human detection on NVIDIA Jetson embedded platform such as PedNet, multipied, SSD MobileNet V1, SSD MobileNet V2, and SSD inception V2. The findings showed that, when compared to other models in the proposed video datasets that have various settings, the SSD MobileNet V2 model offered the best accuracy with the lowest computing time. Reference [94] examined the influence of the aerial image height on the detection's quality. ACF, YOLO MobileNet, and YOLO ResNet50 were evaluated in this work utilizing a separate collection of aerial images that varied in height at 10m, 20m, and 30m. YOLO MobileNet provided the best detection accuracy using 20m height training data. Reference [164] performed a complete performance analysis of a number of cutting-edge object detectors on the MOBDrone data, providing benchmarks for more investigation. Reference [128] examined the accuracy of four cutting-edge object detectors: Faster RCNN, Retinanet, Efficientdet, and YOLOv5 on the "VisBuoy" dataset to determine the best detector for this mission.

V. DATASETS

In ML-based human detection, the choice of aerial training datasets is critical [92]. Many datasets had been proposed and collected in the last decade that include images or videos for humans from UAVs at different heights in different environments and conditions. The most widely used dataset for training the neural networks is the HERIDAL dataset [90].

TABLE 9. Summary of the literature that used SSD models.

Article	Images type	Training Dataset	Model	Processing	Evaluation
[148]	RGB	Not listed	MobileNet SSD model	GS	Same dataset
[111]	RGB	HERIDAL	SSD	Onboard	HERIDAL
[146]	RGB	PASCAL VOC2012	SSD	Onboard	real flight tests.
[124]	RGB	New dataset	Faster R-CNN and SSD-Resnet	Onboard	Same datasets
[141]	RGB	New dataset	new approach influenced by the pyramidal feature extraction of SSD.	GS	Same dataset

HERIDAL was gathered in Bosnia's mountainous regions along the Mediterranean and Sub-Mediterranean Seas. Photo patches are photoshopped images of real individuals in scenarios like crowded public spaces or a stadium. Each image that was recorded for training and testing had a resolution of 4000 by 3000 pixels. Since the HERIDAL collection was primarily created for SAR operations, all the tagged images are solely for humans. The second most known dataset is COCO dataset [166]. COCO is a huge object detection, classification, and labeling dataset that includes 330K photos, 1.5 million object examples, 80 object types, and 91 stuff types. Reference [164] made the MOBDron benchmark available to the public. It is a collection of over 125K drone-view images captured in a marine environment at various heights, angles, and lighting levels. Table 12 presents available datasets of aerial images and videos for SAR operations.

VI. PROCESSING CAPABILITIES

Offline and onboard detection techniques are two alternative ways to find people in aerial photos. In offline techniques, a ML algorithm should be trained on datasets of well-annotated aerial images, then the new images collected by the UAVs can be processed on the algorithm in ground stations. For on-board detection techniques, the algorithm is embedded in a single-board processor (SBP) to analyze the new photos in real-time and sent the processed data to the ground station. Several types of SBP had emerged in the last a few years and made it easier to process the aerial images onboard via hardware such as NVIDIA Jetson Nano, NVIDIA Jetson TX2, NVIDIA Jetson Xavier NX, and Raspberry PI4. Handful of research was done to compare among these SBPs for the purpose of detecting humans in aerial images. For example, [183] evaluated the output of SBPs including the NVIDIA Jetson Nano, NVIDIA Jetson TX2, and Raspberry PI4 using a CNN algorithm constructed with a dataset of fashion product photos. The goal of the research was to help DL models to process images on SBPs in order to achieve excellent accuracy with less hardware. The Jetson TX2 consumes more energy, however, it processes large datasets faster and more accurate compared to other models. Even though Jetson TX2 is the most expansive, lower cost cannot be achieved with better hardware functionality.

Reference [157] examined the YOLO model's detection process and efficiency on 3 distinct SBCs: the Raspberry Pi 4B (RPi) with Intel Neural Compute Stick2 (NCS2),

NVIDIA Jetson Nano, and NVIDIA Jetson Xavier NX (NCS2). The RPi+NCS2's capabilities are more accommodating to lightweight models. Moreover, the procedure of using Jetson devices is much simpler than running a model on RPi + NCS2. The Jetson Nano is an SBC that compromises performance for price.

NVIDIA Jetson TX2 is the most widely used SBP for identifying humans from aerial images in SAR operations. Reference [10] also suggested a fully autonomous UAV equipped with a real-time people detecting system. The system used the YOLOv3 DL approach to detect swimmers in open water using the Nvidia Jetson TX1 embedded hardware. The novel aspect of the proposed system is its integration of Global Navigation Satellite System (GNSS) techniques with CV techniques to enable precise human identification and SAR operation implementation. Reference [163] reviewed several DL algorithms for real-time human detection on NVIDIA Jetson embedded platform such as PedNet, multipeed, SSD MobileNet V1, SSD MobileNet V2, and SSD inception V2. The testing findings demonstrated that, when compared to other models in the proposed video datasets at different settings, the SSD MobileNet V2 model offered the highest accuracy with the lowest computing time. Reference [85] investigated adding processing unit, Jetson Xavier NX, to run emergent AI-based detectors to UAVs. The suggested system was able to send the correct coordinates to the ground station and detect people in real time with acceptable accuracy.

VII. CHALLENGES, RECOMMENDATIONS, AND FUTURE DIRECTIONS

This section discusses the primary challenges in constructing an autonomous UAV system for human detection in SAR missions, and highlights gaps in the literature that hinder the development of effective, accurate, and efficient systems. It additionally offers recommendations on overcoming these challenges and presents future research directions to develop a fully autonomous UAV system that can effectively, efficiently, and accurately detect humans in SAR missions.

- Datasets are not specialized to disaster SAR

One of the main challenges in building accurate and efficient detection systems is the availability of a comprehensive, high-quality, and realistic datasets. Although numerous datasets are available, most of them do not originate from real disasters or disaster-like scenarios. Many researchers agree that having a comprehensive and high-quality dataset is one of

TABLE 10. Summary of the literature that used or proposed other models.

Article	Images type	Training Dataset	Model	Processing	Evaluation
[65]	RGB	New	color signature algorithm	Onboard	Same dataset and Field experiments
[87]	RGB	Two datasets: 1) images taken from a video on the internet. 2) 5 new videos	CNN + SVM	GS	Same dataset
[81]	RGB	New	1. Ring Model 2. Mobility Model 3. Nested k-means method	GS	Field experiments, particularly under time restrictions and harsh weather conditions
[52]	RGB and Thermal	New	New image processing algorithm	Onboard	Field experiments
[130]	RGB	New	Noval ensemble DL method	GS	Same dataset
[88]	RGB	new dataset with three different terrain types (plain, forest, and sea)	anomaly detection algorithms	GS	Same dataset
[135]	RGB	New	motion detection [162], and Kalman filtering	GS	Sam dataset
[149]	3D-LiDAR sensor	New	New CNN	GS	Same dataset
[82]	RGB	ImageNet ILSVRC2014	GoogLeNet integrated with SVM	GS	Same Dataset
[123]	RGB	New	CNN-SS (proposed) CNN, FCN, YOLOv5 (for comparison)	GS	Simulation, models' results comparison
[90]	RGB	HERIDAL	CNN based on VGG16 architecture	GS	Same Dataset
[92]	RGB	HERIDAL	ensemble learning model based on EfficientDET architecture	GS	Same dataset
[93]	RGB	new dataset and tested on HERIDAL	EfficientDET deep neural network	GS	tested on HERIDAL dataset
[71]	RGB	New	RVAI object detection software	GS	Same Dataset
[54]	RGB and thermal	UAVTech	New advanced image processing technique	GS	UAVTech
[95]	Optical	New dataset	A deep neural network that integrates dilated convolutions with Path Aggregation Network.	Onboard	tested on VisDrone2019
[3]	RGB	New dataset	Histogram of Oriented Gradients features are retrieved, and a Support Vector Machine (SVM) is employed for classification.	GS	Same dataset
[147]	Simulated Environment	IDV50, Freiburg Disaster, PENS Victim Detection Research 2017 (PVDR2017)	New Deep CNN	both	Simulation system was developed
[102]	RGB	HERIDAL	New approach employs sequences of sequential pictures rather than just one static picture.	GS	New collected images
[136]	RGB	New dataset	SVM and HOG	GS	real flights
[103]	RGB	New dataset	RGB based spectral signature identification	GS	Compared computer assisted to manual identification.
[107]	RGB	HERIDAL	A new multimodal method	GS	HERIDAL
[108]	RGB	INRIA	Sector Ring Histogram of Oriented Gradients (SRHOG) and SVM	GS	New images
[78]	RGB	New dataset	anomaly detection algorithm	Onboard	Same dataset
[131]	RGB	New	Gaussian Mixture Model & Fourier Transforms	GS	Same dataset
[132]	RGB	New dataset	online ML tool offered by Microsoft Azure (customvisiononline.ai)	GS	Same dataset
[119]	RGB	HERIDAL and new images	deep CNN, called ScoreMap to ROI.	Onboard	Same datasets

the main challenges in building accurate and efficient detectors. For instance, [22] argued that the lack of appropriate quality and quantity training datasets prohibit ML approaches from being more accurate. The authors also claimed that the available datasets are sparse and do not represent a wide

range of real-world scenario conditions. Reference [129] suggested that to develop more reliable detection approaches, it is crucial to gather more comprehensive data covering complicated and unfavorable marine circumstances. Reference [33] suggested that the quality of a dataset should

TABLE 11. Summary of the literature that compare between different models.

Article	Images type	Training Dataset	Model	Processing	Evaluation
[128]	RGB	New dataset VisBuoy	Faster RCNN, YOLOV5, Efficientdet, Retinanet	GS	Same dataset
[89]	RGB	VisDrone, SARD	Faster R-CNN, YOLOv4, RetinaNet, and Cascade R-CNN	GS	New dataset
[142]	RGB	[165]	HOG, YOLO v3, and Mask RCNN	Onboard	Same dataset
[80]	RGB	New dataset	R-CNN, YOLO, and SSD	Onboard	Same dataset and real experiment
[75]	Simulated Environment	UCF-ARG	SVM , AdaBoost, HOG, SSD, Faster R- CNN, YOLO	Onboard	Same dataset
[106]	RGB	New dataset	FPN, Faster R-CNN, Cascade R-CNN, Sparse R-CNN, CornerNet, RetinaNet, Deformable DETR, YOLO-F, DDOD	GS	Same dataset, and compared the new dataset with VisDrone and Okutama
[113]	RGB	HERIDAL and a proprietary video stream	YOLO v5m and Faster R-CNN	Onboard	HERIDAL and a proprietary video stream
[124]	RGB	New dataset	Faster R-CNN and SSD-Resnet	Onboard	Same datasets

be reflected in the diversity of images and their real-world relevance.

Reference [33] who based on their research on urban SAR, also concluded that having a comprehensive dataset is necessary. They suggested that comprehensive datasets should include collapsed building photos from diverse geographical locations and altitudes, level of living, architecture styles, weather conditions, time of day variances, optical camera type, and image features. Same ideas are also applicable to marine SAR where [129] stated that since marine disasters frequently transpire in complicated and unfavorable settings, having comprehensive data in such demanding circumstances becomes crucial. Reference [121] argued that robust identification of maritime rescue targets can be achieved if a representative and diversified dataset is collected to train the detector.

Conditions such as bad weather or poor lighting also affect the detector's performance if they were not considered in the training dataset. Some researchers such as [22] focused on this point in particular where they stated that many conditions and scenarios that arise in real life should be covered by the dataset such as images captured during bad weather. Reference [108] noted that a few postures and poor lighting were the primary limitations to their detection system. The authors suggested extending the training dataset to improve the detection system and overcome its limitations.

Other factors can also affect the comprehensiveness of the dataset, such as the UAV's view angle, human postures, and the number of objects in the image (complexity of the image). Some research has shown the necessity to study such factors. For instance, [108] claimed that it is significant and fascinating to investigate how the UAVs view angle affected the performance of the detectors. Similarly, [184] stated that to achieve a more reliable detection, the training datasets need to include additional noise objects.

Two among of the main factors that affect the dataset quality are the image resolution and the zoom level which have direct impact on the detector accuracy. Reference [87] found it evident in their experiments that the model performed better with increased resolution. This was mostly because

higher quality, more detailed data was obtainable, allowing the decision algorithm to distinguish objects of interest from the background more effectively. On the other hand, because of background objects or some of their portions resemble the items of interest, the authors have noticed a rise in false alarms. Reference [126] also claimed that using cameras with higher resolution will enhance the accuracy of the detection algorithm. Reference [11] suggested investigating various zoom levels of a large UAV's camera to determine the most effective approach for improving image quality without operating large drones across extensive distances. In addition, [125] suggested using better cameras to expand the search area by having the drone fly higher up. Furthermore, [87] found that the height at which pictures were captured also has a major effect on prediction accuracy. The findings indicate that to achieve enhanced detection efficiency, scanning the debris at a lower altitude is preferred.

Some researchers tried to train a model on different datasets simultaneously. For instance, [117] trained a model on different datasets but found that the detection accuracy became worse whereas the model became more general. The authors showed that merging different datasets produced the lowest outcomes since the model was unable to properly adjust to the relevant data in that scenario. However, they claimed that a broader model can be achieved by increasing the amount of training data, and that the order of the sets during training matters, and so does whether the training happens with all images simultaneously or separately on each set.

- Onboard processing

Traditionally, UAVs have been used for disaster SAR missions by being launched, taking pictures or videos, and transmitting the information to a ground control center or cloud. The most significant weaknesses of this strategy are latency and communication medium difference [33]. Modern UAVs are constructed with more advanced on-board processors that offer an option to process images immediately while the UAV scanning the zones. There are some advantages for using onboard processor such as immediate analysis and detection, much lower bandwidth requirements (no need to transfer large amounts of data) and allowing an active

TABLE 12. Available datasets of aerial images and videos for SAR operations.

Dataset	Reference	Type	Number of	Application	Main task	Specialized to SAR?
HERIDAL	[90]	Images	~68K	Mountains, wilderness	real individuals in different scenarios	Yes
COCO	[166]	Images	>330K	General	object detection, classification, and labeling	No
Visdrone	[167]	Images & videos	>10K	General	surveillance, traffic management, and crowded locations with lots of people.	Yes
Open MULTIDRONE	[168], [169]	Images & videos	36 different datasets	General	visual detection, and tracking of bicycles, rowboat	No
[89]	[89]	Images	~2K	Mountains, wilderness	humans for SAR missions	Yes
UAV123 dataset	[170]	videos	123	General	object tracking	Yes
ALOV300++	[171]	videos	315	General	tracking in diverse circumstances	Yes
Okutama-Action	[172]	videos	43 minutes	Mountains, wilderness	human detection and tracking	No
UAV-Gesture	[173]	videos	119	Similar background	Control and Gesture Recognition	No
Semantic Drone	[174]	Images	600	urban scenes	person detection.	Yes
VIRAT	[175]	video	29 hours	urban scenes	action recognition	Yes
[140]	[140]	videos & thermal images	3 videos & ~77K thermal images	campus, grass, and beach	survivor detection in SAR missions	Yes
[141]	[141]	images	2K	urban scenes	People action detection in SAR missions	Yes
UCF101	[176]	video	27 hours	General	Human actions	No
J-HMDB	[177]	videos	32k	different environments	action recognition	No
SWIMMERS	[10]	Videos	50	Marine	Human detection	Yes
Campus	[178]	Frames / videos	>11K	Urban, outdoor environment	object tracking	Yes
UAVDT	[179]	videos	10 hours	urban areas	object detection in variety of weather conditions	Yes
[110]	[110]	Images	310	Mountains, Beaches	People detection in two different SAR scenarios	Yes
SeaDroneSee	[120]	frames	54K	Maritime	Detecting Humans in Open Water	Yes
Manipal-UAV	[106]	videos	33	General	person detection in SAR, military, and surveillance	Yes
MOBDrone	[129]	images	125K	Marine Drowning	people overboard rescue	Yes
[88]	[88]	images	NA	plain, forest, and sea	Color Anomaly Detection from UAV images	Yes
[149]	[149]	images	~3.6K	General	UAV mounted 3D LiDAR for human detection	Yes
ILSVRC 2014	[180]	images	~1.2 million	General	object detection	No
Seagull	[181]	images	>150K	sea monitoring and surveillance	object detection	Yes
AFO Dataset	[130]	video	50	maritime	object detection	Yes
HIT-UAV	[182]	infrared thermal	2.8K	Urban scenes	object detection	No
PRE-FALCO	[98]	RGB / Infrared images	For testing	urban and rural settings	SAR operations	Yes
KAIST Dataset	[98]	RGB color + Thermal images	95k	urban	Pedestrian detection	No
Manipal-UAV	[106]	videos	33	General	small object detection for SAR	Yes
[133]	[133]	image	458	maritime	Human detection for SAR	Yes
[124]	[124]	frames	~32K	land and open water	human detection and tracking in a variety of poses instances.	Yes
ResQuad	[78]	images	863	wilderness	detect prone humans	Yes
[112]	[112]	images	~4.1K	wilderness	Huma detection	Yes

vision approach (using the UAV's active detection system to dynamically modify its flying altitude and route). The drawback of this approach is that deploying ML models on UAVs involves resource limitations such as constrained computational power, memory, and battery life. Moreover, real-time inference demands lightweight architectures that can run efficiently on-board. Balancing accuracy and computational efficiency become critical where researchers explore techniques like model quantization, pruning, and hardware acceleration to optimize inference speed while maintaining acceptable accuracy. However, striking the right balance remains a challenge; a large, complex model may achieve high accuracy but consume excessive resources, impacting flight time and mission effectiveness.

On-board computing is desirable for certain UAV missions, particularly those in which connections may affect performance, such as when large amounts of data must be communicated and/or bandwidth is limited. Today, embedded hardware developers are actively focusing on designing powerful miniature computing systems with low power consumption, particularly GPUs [30]. On-board processing is especially challenging because aerial images present unique challenges, in contrast to standard computer vision issues. Furthermore, when implementing DL models on computing platforms integrated into UAVs, consideration needs to be given to the implicit computational cost associated with their utilization [110]. In general, most of the commercial object detection software is too broad to be utilized with on-board devices. Notably, post-processing allows for the utilization of powerful computer machines, so the constraint of computational power becomes less important. More research needs to be done to assess and enhance the object detection algorithm's efficiency during post-processing [65].

In case of using on-board processing, the complexity of ML algorithms should be taken into account since these processors have much lower computational capabilities. This in turn affects the accuracy of the target's detection. Regarding the UAVs' energy consumption, researchers claimed that the majority of a UAVs' energy is usually used to power its engines [11]. Although using ground station can let mini-UAVs operate longer, it is not very efficient. Consequently, it is necessary to integrate on-board processing with approaches for lowering the energy consumed by UAVs' engines in order to achieve a high degree of energy efficiency [185].

In conclusion, detecting humans from aerial images require very complex ML algorithms which require high computational power. On-board microprocessors can provide immediate processing, but this comes at the expense of detection accuracy and UAV energy consumption. Therefore, it is challenging to build tiny ML algorithms that can provide high detection accuracy.

- General vs. specialized models

As was discussed previously, disasters and their effect differ, and they could occur in numerous environments and under many scenarios. Therefore, the question here is whether it is

practical or beneficial to develop a general model that can be used for all disaster types and environments. Although having such general model provides a flexible and scalable solution, it is very challenging for several reasons. First, it requires a comprehensive and huge dataset collected under different conditions, scenarios, and environments. Second, a large and complex ML model will result from using such huge training data, which will in turn require more computational time and power. Third, having such large model makes it more difficult to use on-board processing.

The choice between a general or specific model for object detection in disaster management is a tradeoff. While a general model offers scalability, it may not be the most efficient for on-board processing due to its computational demands. Conversely, specialized models tailored to specific disaster types and environments can provide accurate detection while minimizing resource requirements. Given the current state of technology, it is more practical to develop disaster-specific and environment-specific detection models. As technology evolves, we anticipate striking a better balance between generality and efficiency. Furthermore, considering the rapid selection needed during disasters, we propose having a suite of models that can be loaded on UAVs based on the disaster type. This would allow users to quickly select the type of disaster and load the most suitable model, enhancing the efficiency of disaster response systems.

- Real implementation

Although there are some research did implement hardware and software systems and performed experiments, these experiments were mostly carried out in controlled environments. As a result, numerous factors pertaining to UAV, detection model, data transmission, and connectivity, including wind speed, lighting, vibration, disruption, and bad weather, have not been sufficiently considered.

SAR scenarios involve constantly changing environments. Drones must navigate through rugged terrains, dense forests, urban areas, and bodies of water. These dynamic conditions introduce challenges for ML models. The same model that performs well in one environment may struggle in another due to variations in lighting, weather, and terrain. For instance, detecting a missing person in a dense forest requires different visual cues than identifying a fire spot in an open field. Ensuring that the ML model adapts to these diverse conditions is crucial but complex. Additionally, drones themselves experience rapid motion during flight, leading to shaky images. Handling this instability while maintaining accurate detection poses a significant hurdle.

To conclude, implementing a drone system for SAR missions involves addressing technical, operational, and logistical complexities. Researchers and practitioners are encouraged to continue to innovate, seeking solutions that enhance the effectiveness of UAV-assisted SAR operations.

- Using both thermal and RGB images

In the realm of imaging technologies, two distinct worlds collide: thermal and RGB (Red-Green-Blue) images. Each spectrum brings its own set of advantages and limitations,

catering to diverse applications. Thermal images, captured based on temperature variations, reveal a hidden world of heat signatures. In contrast, RGB images rely on visible light, painting scenes with vibrant colors and intricate details. Here we explore the differences between these two realms and suggest integrating both for SAR missions. On the one hand, thermal imaging has some advantages such as visibility in low light conditions, penetration through obstacles, and detection of warm objects. However, it has some limitations when the images are taken from a distance [135] and warm conditions [79]. On the other hand, RGB imaging can provide high resolution-colored images that can provide rich visual details, work well in warm weather, and from far distances. However, RGB imaging depends on the visible light, does not penetrate obstacles, and may not work well in bad weather such as heavy rain. Therefore, integrating both forms of imaging can be very advantageous in SAR missions.

Some researchers agreed on the usefulness of utilizing both types of imaging such as [126] who stated that using thermal cameras and infrared technology will improve visibility under low-light conditions. Reference [83] intended to examine the potential for creating hyperspectral images with a UAV that is equipped with a combination of infrared thermal and visible light camera as this would enable them to encode the temperature of a specific region of the picture into the input data. Other articles suggested further investigation of utilizing both imaging technologies such as in [101], [114], [146], and [186].

- The use of multi-UAV cooperative platforms

Cooperative multi-UAV platforms are essential to wide-area SAR missions. These systems use several UAVs to effectively detect victims, geo-localize them, and deliver aid supplies to them in complex areas and environments. Utilizing many UAVs has many benefits including but not limited to 1) reducing the SAR time [184], [187]. 2) enhance the detection accuracy by scanning from different angles. 3) can be used to accurately geolocate humans (based on several detection angles) when conventional GPS methods are not feasible [184]. Even though there was some research proposing the use of multi-UAV cooperative platform such as in [76] and [188], but there are still many opened questions for integrating, coordinating, and optimizing these multi-UAV to realize the possible benefits.

- Enhancing the current approaches

According to the reviewed literature, extensive research efforts have been devoted to building and enhancing object detection approaches for aerial SAR. However, despite significant progress, several limitations persist, leaving room for further advancements. Firstly, existing algorithms for object detection often lack specialized datasets tailored specifically for SAR missions. These datasets should encompass various disaster scenarios, lighting conditions, and the unique challenges posed by UAVs (such as vibration and stability). Without such data, training accurate models becomes difficult. Secondly, computational complexity remains a hurdle. Many state-of-the-art algorithms demand significant

computational power and time, rendering them impractical for on-board deployment. In SAR situations, where timely decisions are crucial, lightweight algorithms are essential. Thirdly, Algorithms for object detection in SAR scenarios should prioritize accuracy over recall. In SAR activities, high precision is essential because false positive detections could make the team's efforts more difficult.

To overcome these challenges, researchers tried to explore lightweight versions of popular object detection algorithms. These models strike a balance between accuracy and efficiency, making them suitable for deployment on resource-constrained UAVs. Additionally, they can be fine-tuned using specialized datasets relevant to disaster SAR scenarios. However, most of these lightweight algorithms need enhancement for better accuracy.

Beyond basic detection, SAR UAVs require additional capabilities. Human tracking, for instance, ensures that once detected, individuals can be followed as they move. Moreover, preprocessing steps—designed to handle varying lighting conditions and mitigate the effects of UAV vibrations—play a vital role in reliable detection. Some researchers suggested using temporal CNNs which allow the ability to integrate many successive video frames [130] and a cooperative multi-neuron network system to recognize people from photos at longer ranges [113].

To sum up, enhancing ML algorithms for object detection in aerial images involves a delicate dance between accuracy, efficiency, and specialized training. By addressing these aspects, we can empower UAVs to play a more effective role in critical SAR missions, ultimately saving lives by minimizing response times.

- Ethical considerations of using UAVs for SAR operations

The use of UAVs in SAR operations raises various ethical concerns that must be considered to ensure responsible use. Several researchers discussed the ethical concerns of using UAVs in SAR missions such as [189] and [190]. A key concern is the privacy and dignity of the people being rescued. UAVs equipped with high-resolution cameras may inadvertently collect sensitive information, resulting in possible privacy violations. It is critical to ensure that data obtained during SAR activities is kept strictly secret and used only for the mission's purposes [190]. Additionally, operators must be educated to respect persons' privacy and avoid undesirable surveillance.

Another major ethical concern is the safety and reliability of UAVs. While UAVs may access hazardous regions where human rescuers cannot, their deployment is not without risks. Technical faults or operator errors can cause incidents, possibly harming both the UAV and people on the ground. Thus, comprehensive testing and adherence to safety measures are required to reduce these dangers. Furthermore, the use of UAVs should be accompanied by strong ethical rules that promote human safety and well-being. Balancing these ethical considerations is essential for the responsible and effective use of UAVs in SAR missions.

VIII. CONCLUSION

The comprehensive review of the literature on human detection from UAV aerial images in SAR scenarios reveals the significant potential of this technology in enhancing the efficiency and effectiveness of disaster response. The integration of UAVs and AI, particularly ML algorithms like deep CNNs, has shown promising results in quickly scanning large disaster areas, identifying and locating victims, and delivering first aid supplies, especially in challenging environments. The presented comparison of methods based on disaster type, environment, case study availability, and UAV system capabilities provides valuable insights into the current state of the field and the areas that require further exploration.

Looking forward, the main challenges to achieving fully autonomous UAV systems for human detection in SAR missions were also discussed in this research. Addressing these challenges will require concerted efforts from researchers, practitioners, and policymakers. The lessons learned from this review and the recommendations provided herein offer a roadmap for future advancements in this critical field for disaster response.

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AHMAD A. BANY ABDELNABI was born in Irbid, Jordan, in 1995. He received the B.S. degree in industrial engineering from Yarmouk University, Jordan, in 2018, and the M.S. degree in industrial engineering from Jordan University of Science and Technology, Jordan, in 2021. He is currently pursuing the Ph.D. degree in modeling and simulation engineering program with the University of Central Florida, Orlando, FL, USA.

From 2018 to 2021, he was a Teaching Assistant with the Industrial Engineering Department, Jordan University of Science and Technology. He also was a Research Assistant with the Engineering Management and Systems Engineering, Old Dominion University, Norfolk, VA, USA, in 2022. Since 2023, he has been a Research Assistant with the IST, School of Modeling, Simulation, and Training, University of Central Florida. He is the author of two peer reviewed journal articles and five conference articles. His research interests include computer vision, operations research, human-centered engineering, digital twins, and supply chain management and logistics.



GHAITH RABADI received the B.S. degree in industrial engineering from Jordan University, Amman, Jordan, and the M.S. and Ph.D. degrees in industrial engineering from the University of Central Florida, Orlando, FL, USA, in 1996 and 1999, respectively.

From 2000 to 2002, he was a Postdoctoral Researcher and a Visiting Assistant Professor with the Department of Industrial Engineering and Management Systems, University of Central Florida. He was an Assistant Professor, an Associate Professor, and a Full Professor with the Department of Engineering Management and Systems Engineering, Old Dominion University, Norfolk, VA, USA, from 2002 to 2008, from 2008 to 2016, from 2016 to 2022, respectively. From 2018 to 2020, he was the Vice President of the Research and Development, Princess Sumaya University for Technology, Amman, Jordan. Since 2022, he has been a Professor and the Graduate Program Director of the School of Modeling, Simulation, and Training, University of Central Florida. He is the author of two books and more than 100 peer reviewed journal articles, conference papers, and book chapters. His research and teaching interests include planning and scheduling, operations research, simulation modeling and digital twins, artificial intelligence, and supply chain management and logistics. Application areas include production systems, disaster response, and transportation system optimization.

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